

EE 66
Fall 2026

Signals, Dynamics, and Information

Discussion 3A

1. Signal Fundamental Frequency

For each of the discrete-time signals below, determine the fundamental period (and corresponding fundamental frequency), or explain why the signal is not periodic.

- (a) $x[n] = e^{i\frac{\pi}{5}n}$ for all integers n .

Answer: We want to find the smallest positive integer p such that $x[n+p] = x[n]$ for all integers n .

$$x[n+p] = e^{i\frac{\pi}{5}(n+p)} = e^{i\frac{\pi}{5}n} e^{i\frac{\pi}{5}p} = e^{i\frac{\pi}{5}n} = x[n].$$

Canceling the common exponentials involving n from each side, we have

$$e^{i\frac{\pi}{5}p} = 1.$$

Therefore, we must have

$$\frac{\pi}{5}p = 2\pi\ell$$

where ℓ is the smallest positive integer ℓ that leads to a positive integer value for p . We have

$$p = 10\ell,$$

so we choose $\ell = 1$, which leads to the fundamental period

$$p = 10$$

and corresponding fundamental frequency

$$\omega_0 = \frac{2\pi}{p} = \frac{2\pi}{10} = \frac{\pi}{5}.$$

- (b) $x[n] = \cos\left(\frac{3\pi}{5}n\right)$ for all integers n .

Answer: We want to find the smallest positive integer p such that $x[n+p] = x[n]$ for all integers n .

$$\begin{aligned} x[n+p] &= \cos\left(\frac{3\pi}{5}(n+p)\right) \\ &= \cos\left(\frac{3\pi}{5}n + \frac{3\pi}{5}p\right) = \cos\left(\frac{3\pi}{5}n\right) = x[n]. \end{aligned}$$

Therefore, we must have

$$\frac{3\pi}{5}p = 2\pi\ell$$

where we seek the smallest positive integer ℓ that leads to a positive integer value for p . We have

$$p = \frac{10\ell}{3},$$

so we choose $\ell = 3$, which leads to the fundamental period

$$p = 10$$

and corresponding fundamental frequency

$$\omega_0 = \frac{2\pi}{p} = \frac{2\pi}{10} = \frac{\pi}{5}.$$

- (c) $x[n] = \sin(n)$ for all integers n .

Answer: This sinusoid is *not* periodic, because no integer p can be found such that $x[n+p] = x[n]$ for all integers n . For a sinusoid x described by, say, $x[n] = \sin(\omega_x n)$ to be periodic, ω_x must be a rational multiple of π . In our case, $\omega_x = 1$, which is not a rational multiple of π . So, x is not periodic.

This can be seen as follows. We want to find the smallest positive integer p such that $x[n+p] = x[n]$ for all integers n .

$$x[n+p] = \sin\left(\omega_x(n+p)\right) = \sin\left(\omega_x n + \omega_x p\right) = \sin(\omega_x n) = x[n].$$

Therefore, we must have

$$\omega_x p = 2\pi\ell$$

where we seek the smallest positive integer ℓ that leads to a positive integer value for p . We have

$$p = \frac{2\pi}{\omega_x} \ell.$$

Since $\omega_x = 1$, we have

$$p = 2\pi\ell.$$

Clearly, no integer value of ℓ exists that can make p an integer. So, the signal is not periodic. For the signal to be periodic, we must have ω_x to be a rational multiple of π , so it cancels the π in the numerator. In other words, if

$$\omega_x = \frac{M}{N} \pi,$$

where M and N are coprime (have no factors in common), then

$$p = \frac{2\pi}{\omega_x} \ell = \frac{2\pi}{\frac{M}{N}\pi} \ell = \frac{2N}{M} \ell.$$

We then select the smallest positive integer ℓ that makes p an integer.

2. Orthogonal Building Blocks

For an integer $k \in \{0, 1, 2\}$, define the vector $\phi_k \in \mathbb{C}^3$ by

$$\phi_k[n] = e^{-ik\frac{2\pi}{3}n}, \quad n = 0, 1, 2.$$

Recall the inner product of two such vectors is $\langle a, b \rangle = \sum_{n=0}^2 a[n] b^*[n]$.

- (a) Write out the 3 entries of ϕ_0, ϕ_1, ϕ_2 , then compute $\langle \phi_0, \phi_1 \rangle$ and $\langle \phi_1, \phi_2 \rangle$. What do you notice?

Hint: Euler's formula, $e^{i\theta} = \cos \theta + i \sin \theta$, will help you evaluate each entry.

Answer: By Euler's formula, $e^{-i\theta} = \cos \theta - i \sin \theta$, so

$$\phi_1[1] = e^{-i\frac{2\pi}{3}} = -\frac{1}{2} - i\frac{\sqrt{3}}{2}, \quad \phi_1[2] = e^{-i\frac{4\pi}{3}} = -\frac{1}{2} + i\frac{\sqrt{3}}{2},$$

and, using $e^{-i\frac{8\pi}{3}} = e^{-i\frac{8\pi}{3} + i2\pi} = e^{-i\frac{2\pi}{3}}$,

$$\phi_2[1] = e^{-i\frac{4\pi}{3}} = -\frac{1}{2} + i\frac{\sqrt{3}}{2}, \quad \phi_2[2] = e^{-i\frac{8\pi}{3}} = -\frac{1}{2} - i\frac{\sqrt{3}}{2}.$$

So

$$\phi_0 = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \quad \phi_1 = \begin{bmatrix} 1 \\ -\frac{1}{2} - i\frac{\sqrt{3}}{2} \\ -\frac{1}{2} + i\frac{\sqrt{3}}{2} \end{bmatrix}, \quad \phi_2 = \begin{bmatrix} 1 \\ -\frac{1}{2} + i\frac{\sqrt{3}}{2} \\ -\frac{1}{2} - i\frac{\sqrt{3}}{2} \end{bmatrix}.$$

Now we compute the inner products:

$$\begin{aligned} \langle \phi_0, \phi_1 \rangle &= (1)(1) + (1) \left(-\frac{1}{2} - i\frac{\sqrt{3}}{2} \right)^* + (1) \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2} \right)^* \\ &= 1 + \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2} \right) + \left(-\frac{1}{2} - i\frac{\sqrt{3}}{2} \right) \\ &= \left(1 - \frac{1}{2} - \frac{1}{2} \right) + i \left(\frac{\sqrt{3}}{2} - \frac{\sqrt{3}}{2} \right) = 0. \end{aligned}$$

$$\begin{aligned} \langle \phi_1, \phi_2 \rangle &= (1)(1) + \left(-\frac{1}{2} - i\frac{\sqrt{3}}{2} \right) \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2} \right)^* + \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2} \right) \left(-\frac{1}{2} - i\frac{\sqrt{3}}{2} \right)^* \\ &= 1 + \left(-\frac{1}{2} - i\frac{\sqrt{3}}{2} \right) \left(-\frac{1}{2} - i\frac{\sqrt{3}}{2} \right) + \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2} \right) \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2} \right) \\ &= 1 + \left(\frac{1}{4} + i\frac{\sqrt{3}}{4} + i\frac{\sqrt{3}}{4} - \frac{3}{4} \right) + \left(\frac{1}{4} - i\frac{\sqrt{3}}{4} - i\frac{\sqrt{3}}{4} - \frac{3}{4} \right) \\ &= 1 + \left(-\frac{1}{2} + i\frac{\sqrt{3}}{2} \right) + \left(-\frac{1}{2} - i\frac{\sqrt{3}}{2} \right) = 0. \end{aligned}$$

Both are zero. (One can check $\langle \phi_0, \phi_2 \rangle$ is zero the same way.)

- (b) Explain why ϕ_0, ϕ_1, ϕ_2 can be used to construct *any* vector in $\mathbb{C}^3$ as a linear combination of them.

Answer: From part (a), ϕ_0, ϕ_1, ϕ_2 are 3 nonzero vectors that are mutually orthogonal. Nonzero orthogonal vectors are always linearly independent, so these 3 vectors are linearly independent. Since $\mathbb{C}^3$ is 3-dimensional, any 3 linearly independent vectors already span the whole space. So $\{\phi_0, \phi_1, \phi_2\}$ spans $\mathbb{C}^3$, meaning every vector in $\mathbb{C}^3$ can be written as some linear combination $c_0\phi_0 + c_1\phi_1 + c_2\phi_2$.

Solution/Answer: Note to TAs: This problem previews the fact (for general N) that these orthogonal complex-exponential vectors form a basis for $\mathbb{C}^N$ — the key fact underlying the Discrete-Time Fourier Series introduced in a later discussion. No need to name this for students yet.